
FIRST SESSION – CONCEPTS AND TOOLS

The first session provides an overview of what AI tools are about and how we can use them in academia.

Concepts and tools to cover

1. What an LLM actually is: prediction, not lookup. Why confident output can still be wrong, and why context matters
2. The landscape: from chatbots to reasoning models to agents, the jagged frontier, and who in the social sciences uses agents
3. Use cases: research, coding, and admin, and why AI rewards expertise
4. How the tools work under the hood: harness, LLM, tools, the agentic loop, context as a budget, prompting habits
5. Choosing your tools: chat vs agentic, Codex vs Claude Code, pricing, models
6. In practice: a prompt for fetching public data through an API, and things to try this week
7. Verifying output: four checks for errors that do not crash
8. Teaching in times of AI: protecting learning, then using AI to teach more
9. Implications for academic work: disclosure, data handling, and what happens to publishing

SECOND SESSION – PRACTICAL WORKFLOWS

The second section is designed to be more interactive. If you have set up Claude Code/Codex through its app or through VS Code, you can use your own materials and run the same prompts.

Topics and workflows that we can cover (order can be changed)

1. Setting up your workspace: VS Code with Codex/Claude Code, `AGENTS.md` / `Claude.md`, Overleaf and Git integration, permissions
2. A referee report on your own paper with `/review-paper` + introduction to skills and subagents
3. Working with sensitive data: what stays on the secure server, and how a codebook and simulated data let the agent write code for it
4. Writing and editing with agentic AI: a `voice.md` built from your own papers, and using AI to edit your draft rather than editing its draft
5. Verifying the paper-code link: checking that the code does what the paper claims, and tracking changes with GitHub or Dropbox
6. Slides from your materials, with a style file that sets the look
7. Making a replication package: README, master script, portable paths